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Project Mid-Defence
on
Ayurvedic Intelligence: Revolutionizing Ayurveda with AI
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ABSTRACT

The project aims to develop an AI-powered platform that integrates modern technology with traditional Ayurvedic knowledge to provide holistic health solutions. The system will utilize image recognition techniques to identify medicinal plants with high accuracy, supported by a curated dataset, and employ Natural Language Processing (NLP) to extract and summarize Ayurvedic remedies based on user symptoms. It will also offer comprehensive information on herbs, human disease analysis, personalized treatment recommendations, and provide Ayurvedic alternatives to modern medicine for ailments. Key features include an AI chatbot, Herbs recognition, "AICare Call" for consultations with Ayurvedic experts, diet planning, and wellness practices such as yoga and meditation, with future plans for virtual yoga training. Leveraging AI models, local storage platforms, the system will be developed in collaboration with Ayurvedic experts, ensuring authentic and accurate recommendations. Ultimately, the project seeks to bridge the gap between modern and traditional healthcare, making Ayurvedic remedies more accessible while preserving and promoting ancient knowledge.

Keywords: *AI-powered platform, Ayurvedic knowledge, Image recognition, Natural Language Processing (NLP), personalized treatment, AICare Call, Wellness practices*

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ACRONYMS AND ABBREVIATION

AI	Ayurvedic Intelligence
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CSS	Cascading Style Sheets
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
JS	JavaScript
ML	Machine Learning
MYSQL	MySQL (Structured Query Language)
NLP	Natural Language Processing

Chapter 1: INTRODUCTION

1.1 Background

Ayurveda, a traditional system of medicine, has long been an integral part of healthcare in Nepal, with a rich heritage of utilizing medicinal plants for various therapeutic purposes. Recent developments in the field of Ayurvedic medicine in Nepal have focused on integrating modern technology with traditional knowledge. This includes the increasing use of AI to enhance the identification and analysis of medicinal plants. AI-based projects, particularly those using machine learning algorithms, can scan herbal plants, identify them, and provide accurate analysis and potential remedies based on centuries of traditional knowledge and modern pharmacological research. This convergence of technology and tradition holds immense promise in improving the accessibility and effectiveness of Ayurvedic treatments (Ripu M Kunwar, 2010).

Recent trends indicate a growing interest in AI-driven systems that can digitize and catalog the diverse medicinal plants of Nepal, many of which are integral to local healing practices. With the increasing exploitation of medicinal plants in Nepal for commercial purposes, especially through the export trade there is a critical need for AI systems to assess plant health, track sustainability, and ensure the preservation of biodiversity. AI technologies could also assist in evaluating the pharmacological potential of these plants, as discussed in various studies on traditional herbal medicine in Nepal, where researchers are keen to discover new applications for these plants in combating modern health issues like COVID-19 (Dipak Khadka, 2021).

However, there are certain drawbacks and challenges that must be addressed. The existing works on AI-based plant identification systems in Nepal often lack a comprehensive database of regional medicinal plants, which limits the AI's ability to offer precise remedies. Additionally, there is a need to integrate ethnobotanical knowledge from diverse ethnic groups in Nepal to enhance the accuracy of AI-generated solutions. Furthermore, while AI-based systems hold significant promise,

the lack of standardization and validation of these remedies against scientific research remains a major hurdle. The growing concerns over the use of fixed-dose herbal combinations, as noted in some studies, also highlight the necessity for a careful approach when developing AI-driven solutions to ensure safety and efficacy. Thus, while AI-based plant identification systems offer an exciting avenue for enhancing Ayurvedic practices in Nepal, addressing these gaps will be crucial for their successful implementation and acceptance (Arjun Poudel, 2016).

1.2 Statement of Problem

The Ayurvedic herbal medicine project faces challenges such as:

1. Limited accessibility to herbal knowledge and Ayurvedic practices.
2. Difficulty in offering personalized health recommendations.
3. Lack of integration between traditional Ayurveda and modern technology.

AI aims to resolve these by using AI for herb identification, NLP for personalized remedies, and Ayurvedic health assessments, combining ancient wisdom with modern tech to provide a reliable and personalized health solution.

1.3 Objectives

The main objectives of the project includes:

1. To provide people with insights into Ayurvedic medicine, its practices, and benefits through an intuitive, easy-to-use interface.
2. To leverage AI to offer personalized recommendations for health, diet, and lifestyle.
3. To provide users with access to consultations, making Ayurveda more accessible through virtual interactions.

1.4 Motivation and Significance

The limitations of traditional Ayurvedic practices, such as limited access to accurate herb identification and personalized health recommendations, motivate the need for modernization. This project aims to integrate AI with Ayurvedic knowledge to enhance accessibility and provide reliable, personalized solutions. By using AI for plant identification, NLP for remedy suggestions, and Ayurvedic health assessments, the system will bridge the gap between ancient wisdom and modern technology. The proposed solution has significant implications for improving Ayurvedic healthcare, making it more accessible, efficient, and personalized for users, while preserving traditional knowledge and practices.

1.5 Scope of Work

The scope of this project includes the following:

1. Development of an AI-powered system for identifying medicinal herbs through image recognition.
2. Integration of Natural Language Processing (NLP) for personalized Ayurvedic remedy suggestions based on user symptoms.
3. Implementation of a user-friendly interface for easy navigation and interaction.
4. Integration of a consultation feature, "AICare Call," to connect users with Ayurvedic experts.

Chapter 2: RELATED WORKS

2.1. Overview of Existing Systems

Existing Ayurvedic health systems like Patanjali, AyuRythm, AIwell, Nepal Ayeveda, and AyuCare offer various features, including product offerings, personalized health solutions, and expert consultations. However, they have limitations such as regional availability, lack of scientific validation, privacy concerns, and technical issues. These gaps highlight the need for a system that combines traditional Ayurvedic knowledge with modern technology, improving accessibility, personalization, and reliability.

2.1.1. Patanjali

Patanjali, founded in 2006 by Baba Ramdev and Acharya Balkrishna, is a well-known Indian brand offering a wide range of Ayurvedic products, including herbal medicines, personal care items, and food products. Known for its focus on natural ingredients and affordability, Patanjali has established itself as a household name with an extensive distribution network across India. The company integrates traditional Ayurveda with modern production techniques to ensure quality and accessibility to consumers globally (Singh, 2016).

2.1.2. AyuRythm

The app is designed to offer personalized Ayurvedic health solutions, enabling users to integrate Ayurveda into their daily lives. It provides tailored recommendations for diet, lifestyle, and wellness practices based on individual doshas (body constitutions). AyuRythm also includes features like access to a wide range of Ayurvedic remedies, consultations with expert practitioners, and educational content on Ayurvedic medicine. This app aims to make Ayurvedic knowledge more accessible, supporting users in achieving a balanced and holistic lifestyle (Monthly Ayurveda Bulletin, 2019).

2.1.3. AIwell

AIwell was launched in 2022 by Ayur.AI as an integrative health app that combines Ayurveda with modern medicine. It offers personalized health assessments and wellness plans based on individual Ayurvedic doshas, integrates wearable devices for real-time health monitoring, and provides access to a network of health experts. AIwell aims to bridge traditional healing practices with modern technological advancements to provide a comprehensive wellness experience (AI Well, n.d.).

2.1.4. Nepal Ayurveda

Nepal Ayurveda is a mobile app developed by TechKharkhana to make Ayurvedic healthcare more accessible to the Nepalese population. It allows users to book appointments with certified Ayurvedic doctors, purchase Ayurvedic products, and access wellness information based on traditional healing practices. The app offers personalized health recommendations, information on Ayurvedic herbs, and a marketplace for purchasing wellness products. While it emphasizes user privacy, some users have reported technical issues and difficulties with navigation, but the app remains a valuable tool for integrating Ayurvedic practices into modern healthcare (Trxrsop, 2024).

2.1.5. AyuCare

Ayurcare is an Ayurvedic wellness center in Dubai, offering a wide range of traditional Ayurvedic treatments such as relaxation therapies, weight loss therapies, detox, skin and hair care, and joint care. They also provide specialized treatments for pregnancy care, eye care, and disease-specific therapies. The center has over two decades of experience in delivering authentic Ayurvedic services and is dedicated to promoting holistic wellness (ayurcare, 2025).

2.2. Comparison of Features

Table 1: Features and Drawbacks

Platform	Features	Drawbacks
AIwell	Health assessments based on Ayurvedic doshas. Integration with wearable devices for monitoring.	Limited regional availability. Privacy concerns with data usage.
Ayurhythm	Rhythm-based Ayurvedic therapies. Personalized guided therapy sessions.	Niche appeal. Limited scientific backing for some therapies.
Patanjali	Wide range of Ayurvedic products. Affordable pricing and strong distribution.	Quality control issues. Limited international availability.
Ayucare	Online Ayurvedic consultations. Personalized health plans.	Dependent on internet connectivity. Generic advice without deep personalization.
Nepal Ayurveda	Provides Ayurvedic consultations and herbal remedies. Focuses on traditional Nepalese healing practices	Limited technological features compared to global platforms. Availability of authentic products can be challenging

2.3. Gaps in Existing Systems

1. Many platforms like AIwell have limited regional availability and raise privacy concerns regarding personal data.
2. Ayurhythm and Ayucare offer niche or generic services, with Ayurhythm lacking scientific validation and Ayucare providing advice that may not be personalized.
3. Patanjali faces quality control issues and limited international availability, while Nepal Ayurveda has limited technological features compared to global platforms.

2.4. Significance of Proposed Work

1. Integration of Tradition and Technology: The system merges Ayurvedic knowledge with modern AI and NLP for personalized health solutions, making traditional practices more accessible.
2. Accurate Plant Identification and Remedies: AI ensures precise identification of medicinal plants, and NLP recommends personalized Ayurvedic treatments based on symptoms.
3. Comprehensive Health Support: The platform offers holistic care with expert consultations, customized diets, and yoga routines, promoting overall well-being

Chapter 3: METHODOLOGY

This AI-based Ayurveda project outlines the different approach, tools, and methods employed to achieve the project's objectives. This section begins with an explanation of the overall strategy, focusing on leveraging artificial intelligence to analyze Ayurvedic medicinal knowledge, identify medicinal plants, and recommend personalized treatments. The design emphasizes the integration of machine learning models, natural language processing for Ayurvedic text interpretation, and computer vision for plant identification. The proposed approach highlights its relevance in preserving traditional knowledge, its effectiveness in providing modern applications, and the innovative aspect of combining Ayurveda with advanced AI technologies.

The project focuses on developing a robust system capable of identifying medicinal plants using computer vision, extracting information from Ayurvedic texts with natural language processing (NLP), and providing personalized treatment recommendations through machine learning models. Key technologies and tools include TensorFlow, and PyTorch for AI and machine learning, OpenCV for computer vision tasks or image processing, Ollama for large volume text analysis & different Natural Language Toolkits and a user-friendly interface developed using different JavaScript Frameworks.

3.1 Requirements Gathering

- **Objective:** The key requirements and goals of the project are identified through a combination of extensive research on Ayurvedic principles, and analysis of user needs. The objectives is documented with a focus on integrating Ayurvedic medicinal knowledge into a modern AI-driven framework, enabling plant identification, personalized treatment recommendations, and promoting awareness of traditional healthcare practices.
- **Techniques Used:** The project utilized Literature Reviews to examine Ayurvedic texts, research papers, and existing digital tools, identifying gaps

and opportunities for innovation in the field. This research provided insights into Ayurvedic practices and the potential for integrating AI-based solutions. Additionally, Data Analysis is employed to analyze publicly available datasets on medicinal plants and user behavior trends related to healthcare apps. This analysis help to inform the development of the system by understanding the relationships between plant properties, treatment effectiveness, and user preferences, ensuring the app meets the needs of its users.

- **Tools Used:** For requirement documentation and collaboration, we used tools like CSV files to organize plant data, symptom-treatment mappings, and availability zones in a structured format. MS Word supported the preparation of formal reports and literature reviews with collaborative editing. GitHub Projects and Issues helped manage development tasks, track progress, and align features with requirements. Additionally, Google Colab is used to prototype and document AI models, combining code, outputs, and notes for effective experimentation.

3.2 System or Model Design

3.2.1 Image Classification

The methodology for the work plant classification involves a series of procedures, as illustrated in Fig.

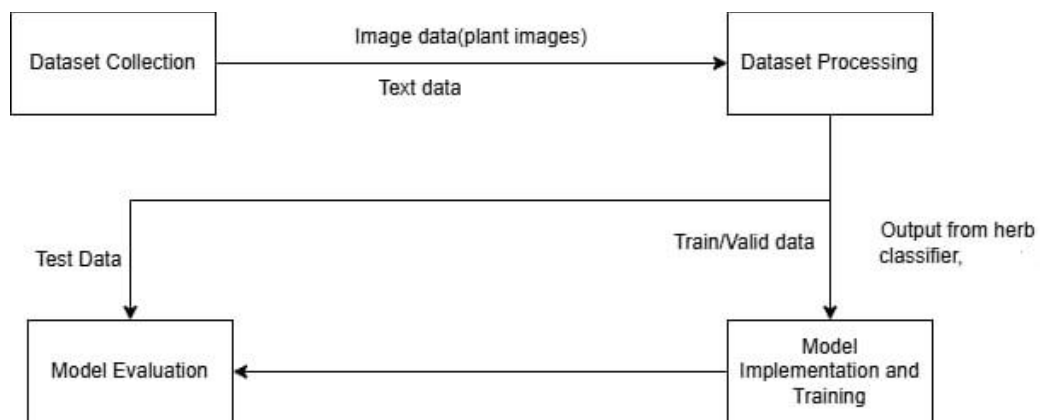


Figure 1: Image Classification

For this project, we utilized the Indian Medicinal Leaves Image Dataset from Kaggle (Shah, 2023), published in 2023. This dataset includes a collection of medicinal plant images captured under various natural conditions. Although the dataset covers a wide range of Indian plants, we specifically focused on filtering and using only those species commonly found in Nepal, such as Neem and Tulsi. The dataset was extracted in a Google Colab environment from a zip file stored in Google Drive. We verified the class folders after extraction to ensure the relevant plant images were properly organized.

After loading the dataset, we applied image preprocessing using torchvision.transforms, including resizing all images to 224×224 pixels and applying random horizontal flips for basic augmentation. We used PyTorch to structure the dataset and created data loaders for batch training. For model implementation, we used ResNet-18 with pretrained weights, modifying its final layer to match the number of selected classes. The model was trained using the Adam optimizer and cross-entropy loss for 20 epochs on GPU. After training, the model was saved for future use. An additional prediction function was developed to test the model on new images, demonstrating its ability to classify medicinal plants relevant to Nepalese biodiversity.

3.2.2 AI powered Chatbot

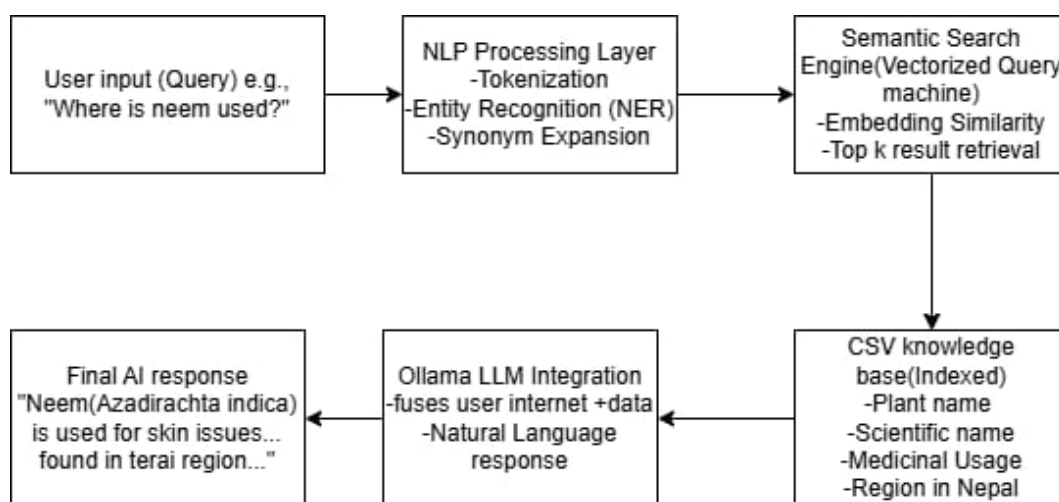


Figure 2: AI powered Chatbot

This project implements an AI-based Chatbot specifically designed to respond to queries about medicinal plants of Nepal. The Chatbot uses the Ollama language model and a structured .csv dataset that contains information such as plant name, scientific name, medicinal usage, and regions in Nepal where the plant is found. When a user inputs a question like “What is the use of Tulsi?”, the Chatbot extracts relevant data from the CSV and generates a meaningful, natural-sounding reply.

Powered by an LLM, the Chatbot understands user questions using Natural Language Processing (NLP) and connects them to the structured plant database. The Ollama model ensures that responses are coherent, contextually appropriate, and relevant to the region-specific data provided. This system demonstrates how AI can be used in healthcare awareness, helping users access traditional plant knowledge through conversational AI.

3.2.3 System Flow Diagram

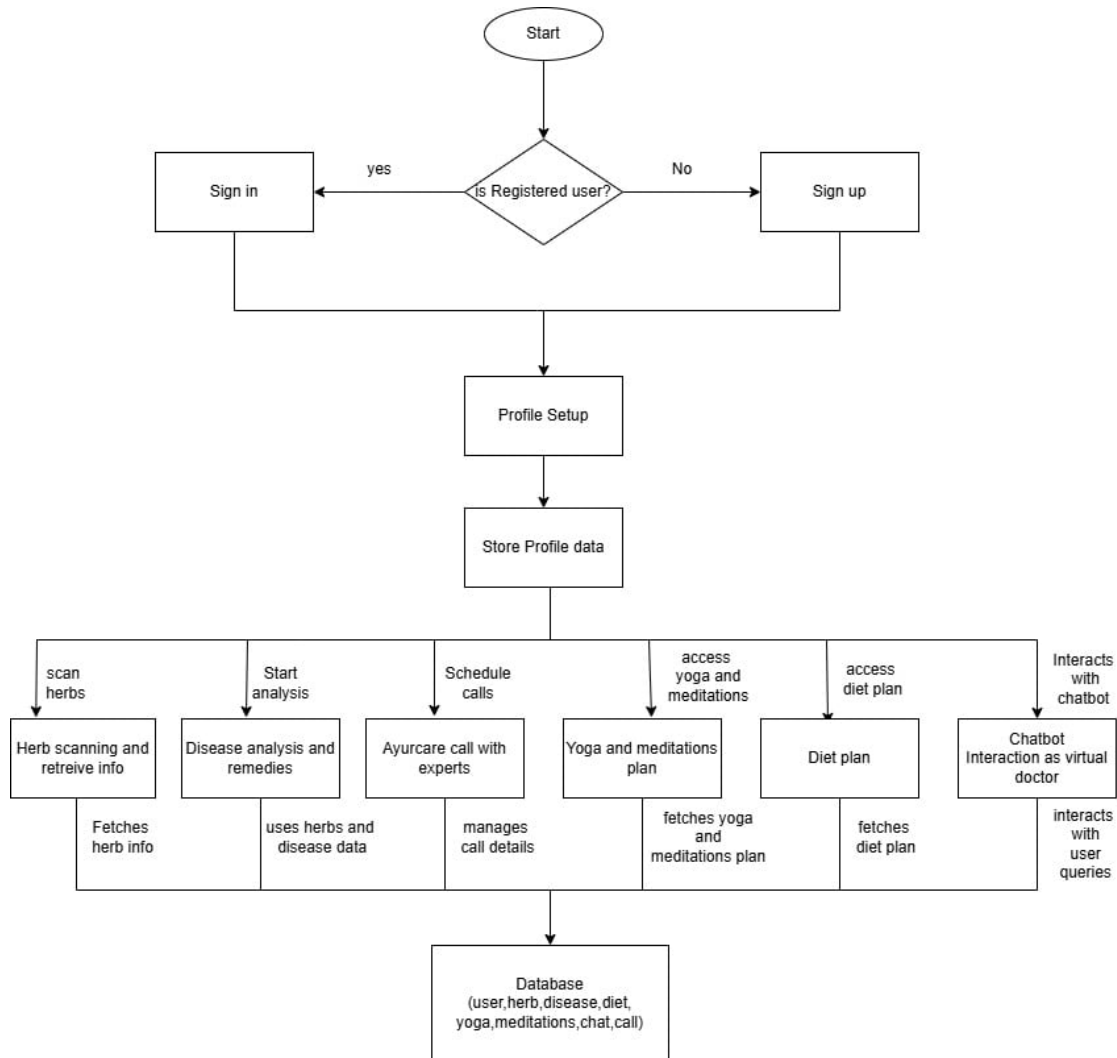


Figure 3: System Flow Diagram

The system flow begins with the user uploading an image of a medicinal plant or entering symptoms into the platform. The uploaded image is processed using computer vision techniques to identify the herb, while the symptoms are analyzed to extract relevant information. Identified herbs and interpreted symptoms are matched with a plant database to generate personalized Ayurvedic treatment suggestions. The results, including plant details, remedies, and wellness tips, are displayed to the user through an intuitive interface. Additional features like the Chatbot and AICare Call offer real-time interaction and expert consultation.

3.2.4 Overall Framework

The system operates through a client-server architecture where the client-side communicates with the server via APIs or GraphQL to ensure efficient data exchange and real-time responsiveness. The server-side manages tasks such as image classification, NLP-based symptom analysis, and generating personalized Ayurvedic recommendations. The AI pipeline begins when the user provides input either an image or symptom-related text which is then preprocessed before being analyzed by AI models for classification and treatment suggestion. The final results are returned to the client in an intuitive format, ensuring smooth and effective interaction.

3.2.5. Core Components

- **Data Management Module:** The Data Management Module handled collecting, organizing, and preprocessing plant images for the classification model. It ensured data was class-wise structured, resized, and augmented for smooth integration into the training pipeline.
- **Feature Engineering:** Feature engineering was performed using a pre-trained ResNet-18 model, which automatically extracted high-level image features for classification, eliminating the need for manual feature design and ensuring consistent input through standardized image resizing.
- **AI Models:** The AI model used is a fine-tuned ResNet-18 CNN, pretrained on ImageNet and adapted to classify Nepalese medicinal plants by replacing its final layer. It was trained using the Adam optimizer and cross-entropy loss for accurate image-based herb identification.
- **User Interaction Module:** The User Interaction Module, built with HTML, CSS, and JavaScript, provides a simple web interface where users can upload plant images, receive classification results, and view related details, enabling smooth communication with the backend AI model.

3.3 Technology Stack

3.3.1 Platform: Google Colab

We used Google Colab as the main development environment for the entire project. It provided free access to GPU hardware, which significantly accelerated the training of our deep learning model. Additionally, Colab's seamless integration with Google Drive allowed us to store and retrieve both the dataset and trained model efficiently.

3.3.2 Storage: Google Drive

Google Drive was used to store the original dataset in .zip format and also to save the trained model file (.pth). We mounted Drive into Colab to enable direct access to files without manual uploads or downloads.

3.3.3 Programming Language: Python

The entire project was implemented in Python, chosen for its simplicity and extensive support for machine learning libraries. Python was used to write code for data loading, preprocessing, model training, evaluation, and prediction.

3.3.4 Deep Learning Framework: PyTorch

We used PyTorch as the core deep learning framework. PyTorch provided the tools needed to define the neural network architecture (ResNet-18), apply loss functions, perform optimization using backpropagation, and manage the training loop. The model was also moved to GPU using PyTorch's device management utilities.

3.3.5 Computer Vision Toolkit: torchvision

The torchvision library was essential for image processing and loading. We used transforms to apply preprocessing like resizing and horizontal flipping. The dataset was organized using datasets.ImageFolder, and the pretrained ResNet-18 model was imported from torchvision.models.

3.3.6 Pretrained Model: ResNet-18

We implemented ResNet-18, a deep convolutional neural network pretrained on ImageNet, for image classification. We replaced its final fully connected layer to match the number of target medicinal plant classes. This use of transfer learning allowed for efficient training on a smaller dataset.

3.3.7 Image Handling: PIL (Python Imaging Library)

For the prediction phase, we used PIL to open and convert new input images to RGB format. These images were then transformed and passed to the trained model to predict the plant class.

3.4. Data Management

Data management involved organizing the dataset into appropriate formats and storing it on a cloud platform for easy access during development. Preprocessing techniques such as resizing, normalization, and data augmentation were applied to prepare the data for training. The dataset was divided into training, validation, and testing subsets to facilitate model evaluation. All processes related to data handling were automated and tracked to ensure reproducibility and efficient workflow.

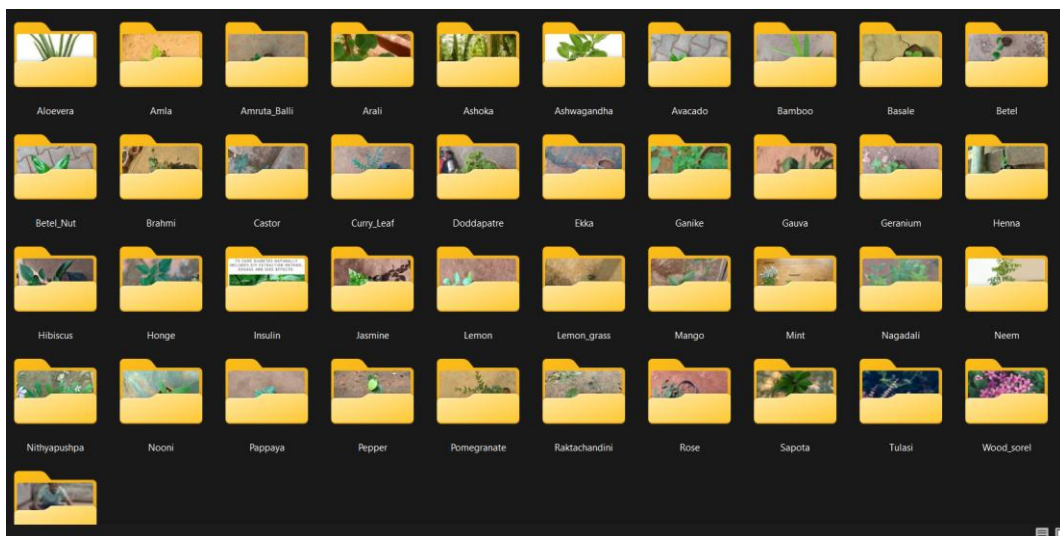


Figure 4: Dataset Used

3.5 Model Evaluation

3.5.1 Training and validation matrices

The model demonstrates a robust learning process and effective performance improvement across all evaluated metrics.

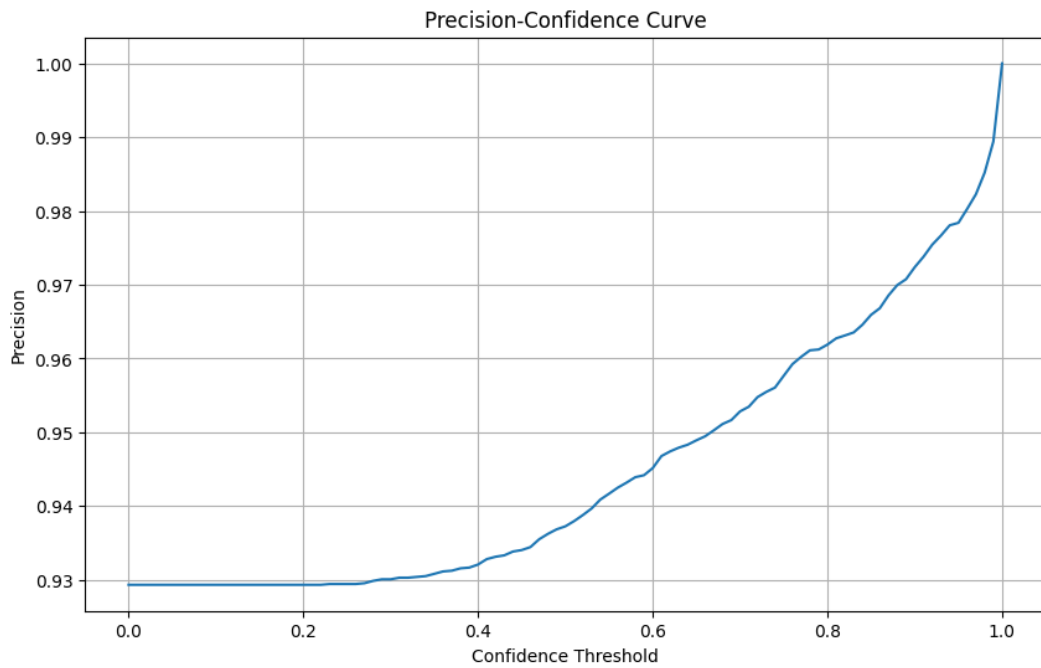


Figure 5: Precision-Confidence Curve

The Precision-Confidence Curve, which illustrates how the precision of a model changes with varying confidence thresholds. As the confidence threshold increases, precision also increases, starting around 0.93 and reaching nearly 1.0. This behavior indicates that the model becomes more selective at higher thresholds, predicting only when it is very confident, which results in fewer false positives and higher precision.

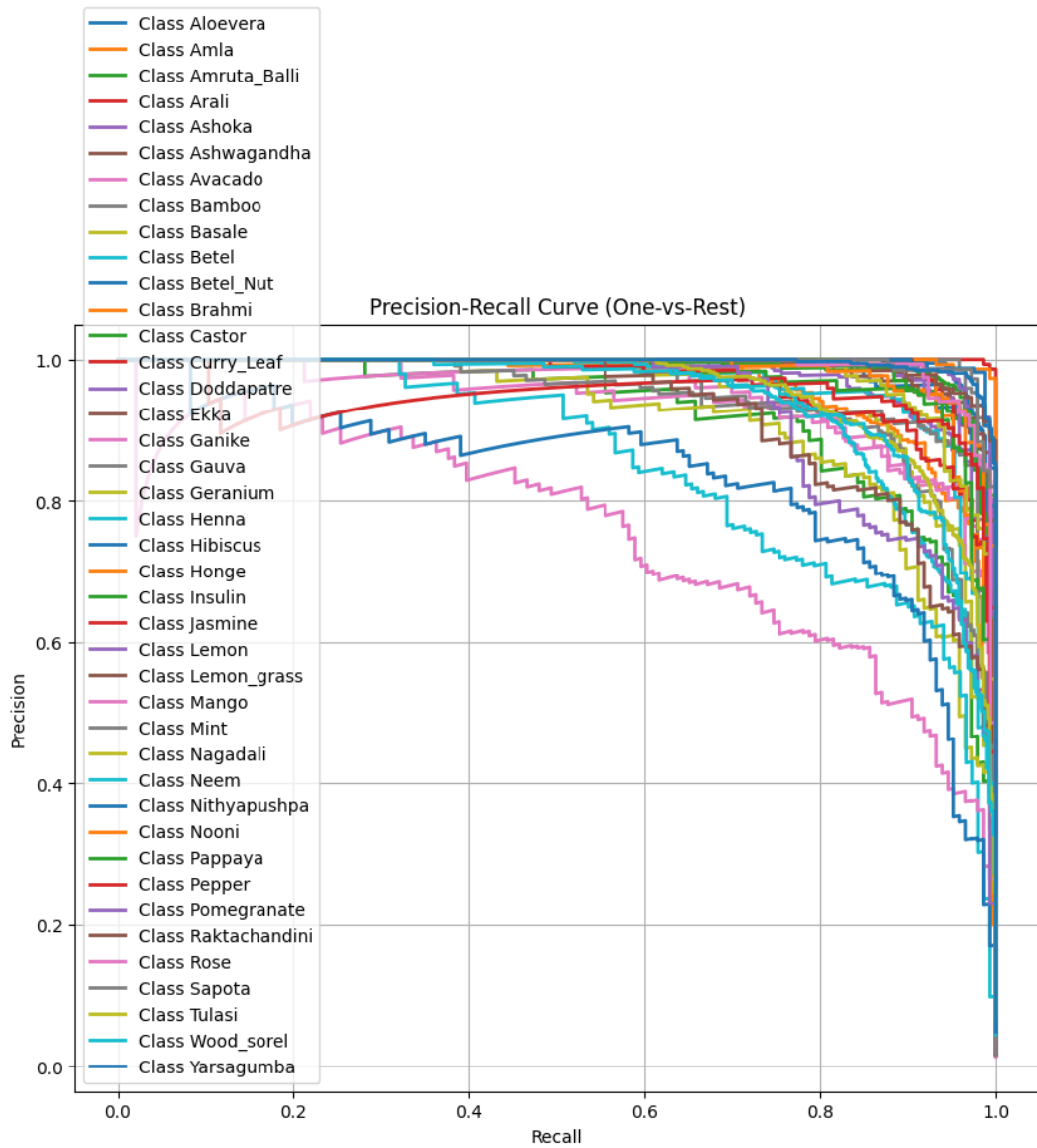


Figure 6: Precision Recall Curve

The precision-recall curve visualizes the trade-off between precision and recall across varying thresholds. A high area under this curve indicates that the model maintains a good balance between correctly identifying relevant plants and minimizing false detections. This performance measure confirms the classifier's robustness in identifying medicinal plants accurately.

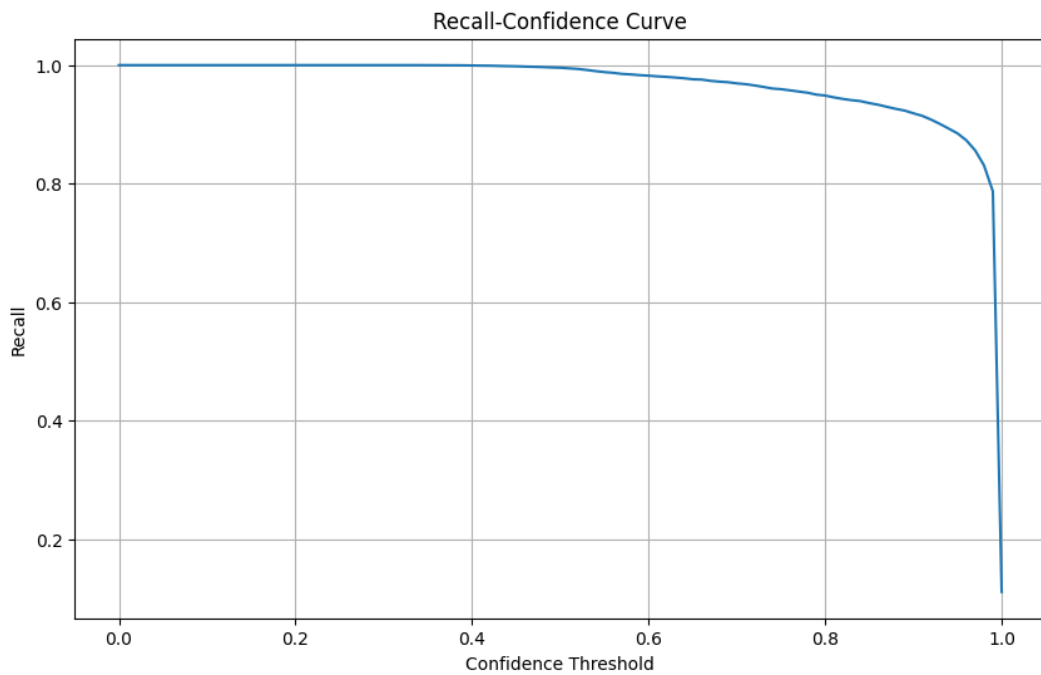


Figure 7: Recall Confidence Curve

The Recall-Confidence Curve, demonstrating how recall varies with the confidence threshold. Recall remains high (near 1.0) for low thresholds, meaning the model captures most true positives. However, as the threshold increases, recall gradually declines and drops sharply near a threshold of 1.0. This shows that while the model becomes more precise, it starts missing more actual positives, reducing its ability to identify all relevant instances.

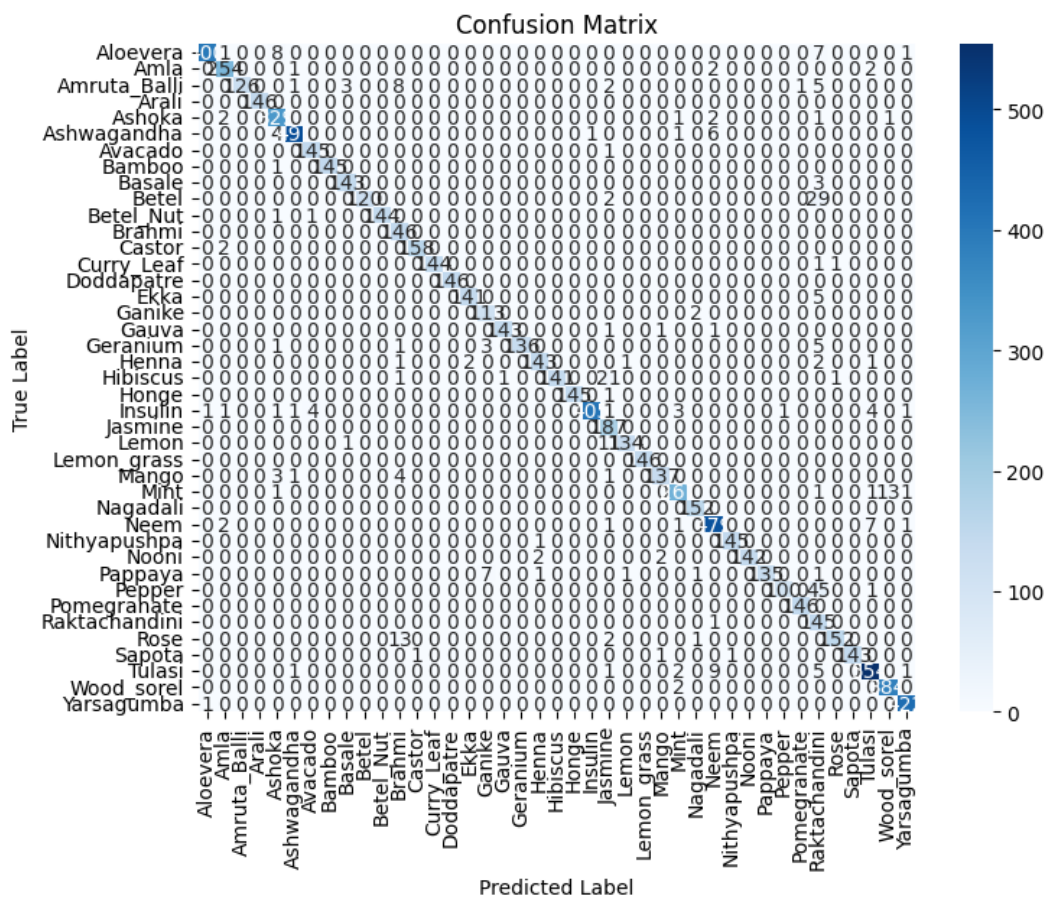


Figure 8: Confusion Matrix

The figure shows that the classifier has high accuracy overall, with most herbs correctly predicted. Misclassifications are minimal and mostly between visually similar plants, which we plan to improve with more data and fine-tuning.”

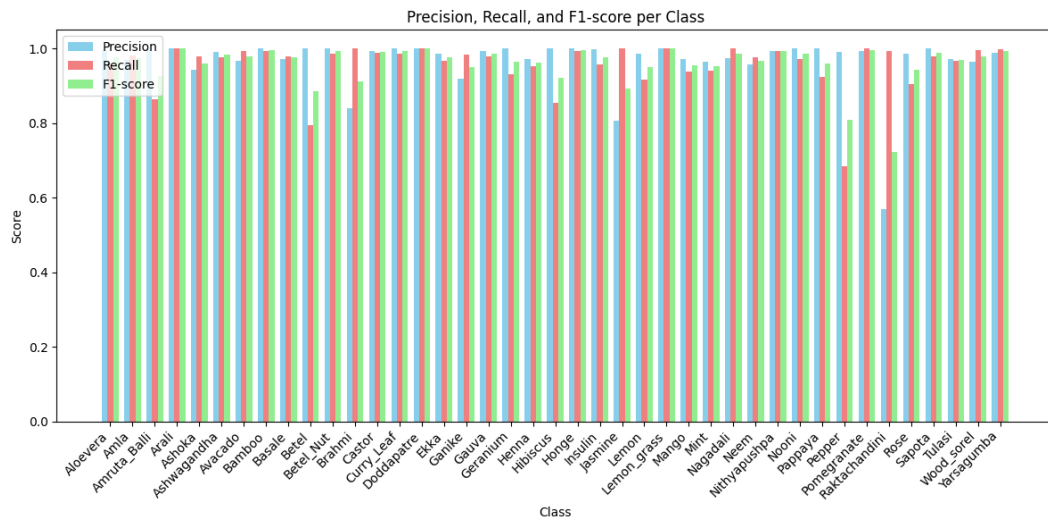


Figure 9: Precision, Recall and F1-score per class

This chart shows the precision, recall, and F1-score for each individual medicinal plant class. It highlights how well the model performs on each specific herb, indicating strengths and weaknesses per class. This information is vital for targeting improvements in underperforming categories.

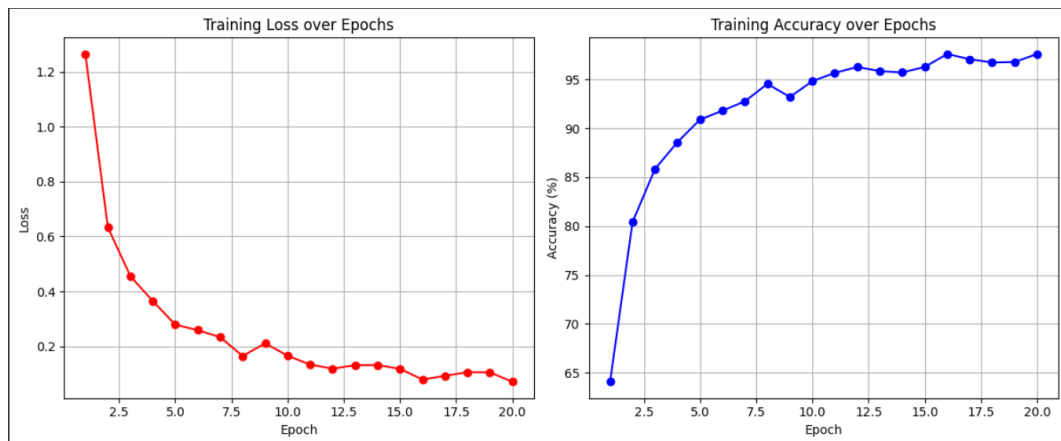


Figure 10: Training Loss and Accuracy over Epochs

In the left graph, training loss starts high (around 1.25) and decreases steadily, dropping below 0.1 by the 20th epoch. This indicates the model is learning effectively and reducing its error over time. The right graph shows training accuracy improving from about 64% to 97%, with most gains occurring in the first 10 epochs and then leveling off.

3.6 Software Development Process

The project adopts Agile Methodology for iterative development.



Figure 11: SDLC (Group Work)

3.6.1. Identify/Research:

In this phase, key information was gathered by reviewing Ayurvedic texts, research papers, and existing digital tools on plant identification and treatments. Public datasets on medicinal plants were also analyzed. This phase helped identify challenges, opportunities, and define the project's scope.

3.6.2. Project Preparation:

The Project Preparation phase established the foundation for successful system development by selecting tools, technologies, and methodologies. This included gathering data from sources like Google Scholar and open-access plant databases, setting up the development environment, organizing resources, and preparing the team. Clear goals, a project timeline, and key milestones were also defined to guide the development process.

3.6.3. Design:

The Design phase centers on creating the system architecture and user interface to ensure a user-friendly experience. For this project, it involved designing accessible web interfaces and planning AI models for plant identification and treatment suggestions. The backend architecture, including databases like MySQL and communication protocols such as APIs and GraphQL, was also designed to ensure scalability and smooth integration with the front end.

3.6.4. Execution:

The Execution phase involved implementing the project plans, including developing AI models, backend setup, front-end development, and tool integration. Python was used for machine learning, JavaScript for the front end, and OpenCV for image processing. AI models for plant identification, NLP for Ayurvedic text analysis, and recommendation systems were trained and tested, with continuous refinement to ensure accuracy before deployment.

3.6.5. Sustain:

The Sustain phase focuses on maintaining the system post-launch. This will include addressing any bugs, improving performance, and ensuring that the system remains up to date with evolving data and user needs. After the initial deployment, the system will be monitored for user feedback, and necessary updates or enhancements will be made, including fine-tuning AI models and integrating new plant data. Regular updates to the software and backend will ensure continued functionality, and additional features or enhancements may be added based on user requests or advancements in AI technology.

3.6.6. Monitor:

The Monitor phase is about continuously evaluating the system's performance and user engagement to ensure it meets expectations and delivers value. During this phase, the application will be monitored for issues such as system downtime, errors,

and user experience concerns. User data will be analyzed to assess the effectiveness of the plant identification models and treatment suggestions. Analytics tools will be employed to track system performance, and any necessary adjustments to the models or interface will be made. Regular monitoring will also help track the system's progress in addressing gaps identified during the research phase and ensure its ongoing improvement.

3.7 Version Control

Git was used for version control to efficiently manage code changes. GitHub served as the central repository, enabling easy collaboration, version tracking, and branching for different features or bug fixes. This provided a history of changes and allowed rollbacks when necessary.

3.8 Collaboration and Task Management

GitHub Projects was utilized for collaboration and task management throughout the project. It helped organize and monitor tasks by visualizing the project timeline, assigning responsibilities, setting priorities, and tracking progress. This tool ensured the team stayed focused on high-priority features, managed sprints efficiently, and maintained clear communication among developers, testers, and stakeholders.

Chapter 4: OUTCOME AND PENDING TASKS

4.1. Expected Outcome

The AI-based system will leverage modern technology to bridge gaps in the accessibility and application of Ayurvedic knowledge while offering a comprehensive health solution. The system will utilize advanced image recognition techniques to identify medicinal plants with robust accuracy, validated through a curated dataset. It will also incorporate Natural Language Processing (NLP) to extract and summarize Ayurvedic remedies based on user-inputted symptoms, providing personalized treatment suggestions. In addition, the platform will offer detailed information on medicinal herbs, analyze diseases, and suggest remedies specific to ailments like digestive disorders, skin conditions, and stress management. A unique feature, "AICare Call," will allow users to connect with Ayurvedic experts for personalized consultations. The system will further enhance user experience with a news feed that provides updates on Ayurvedic practices, trends, and research. Additionally, users will have access to customized diet plans and wellness recommendations, including yoga and meditation routines, with the eventual inclusion of virtual yoga training. By integrating these elements, the system will offer a holistic approach to health, combining Ayurvedic wisdom with modern technology for comprehensive care.

4.2 Achieved Outcomes

4.2.1 Web Application Functionality

Frontend and backend have been successfully developed using modern web technologies. User interfaces for patient login, doctor login, Chatbot interaction, and homepage navigation are fully functional.

4.2.2 Initial Chatbot Implementation

A prototype Chatbot has been integrated for basic symptom-based conversations. It can handle general Ayurvedic queries and guide users through the platform.

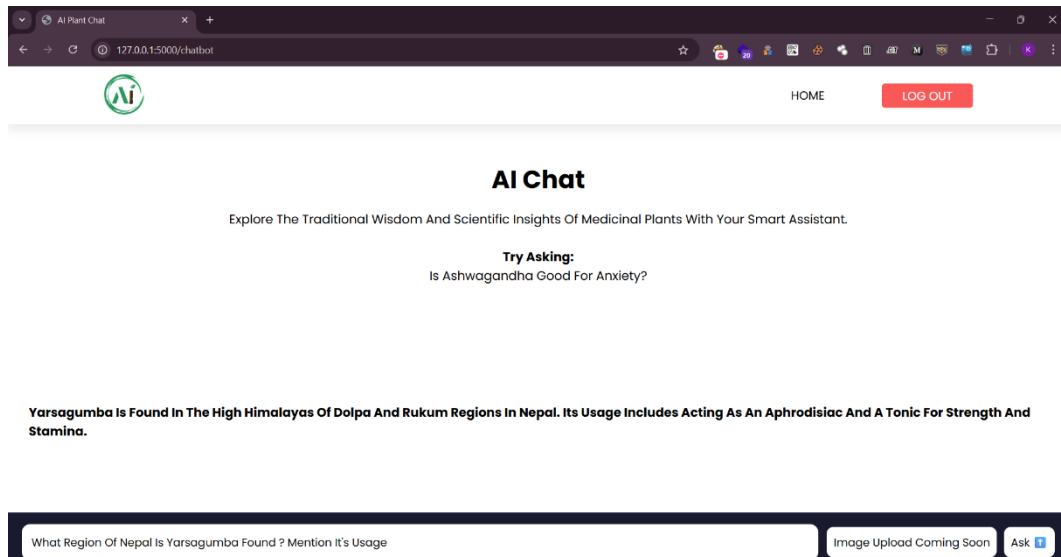


Figure 12: Chatbot

4.2.3 Herbal Image Classifier (Initial Version)

A basic herb image classification model has been implemented. It identifies known herbs from user-uploaded images and returns the plant name as output.

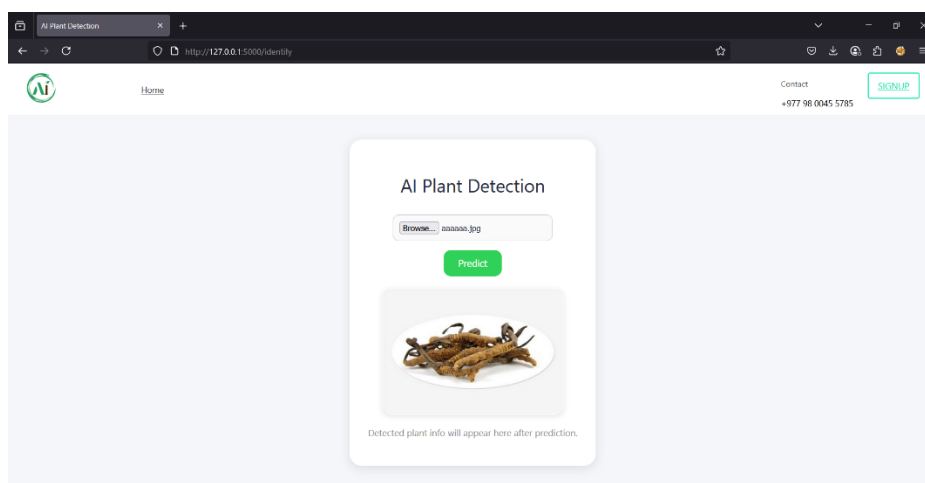


Figure 13: Image Identifier

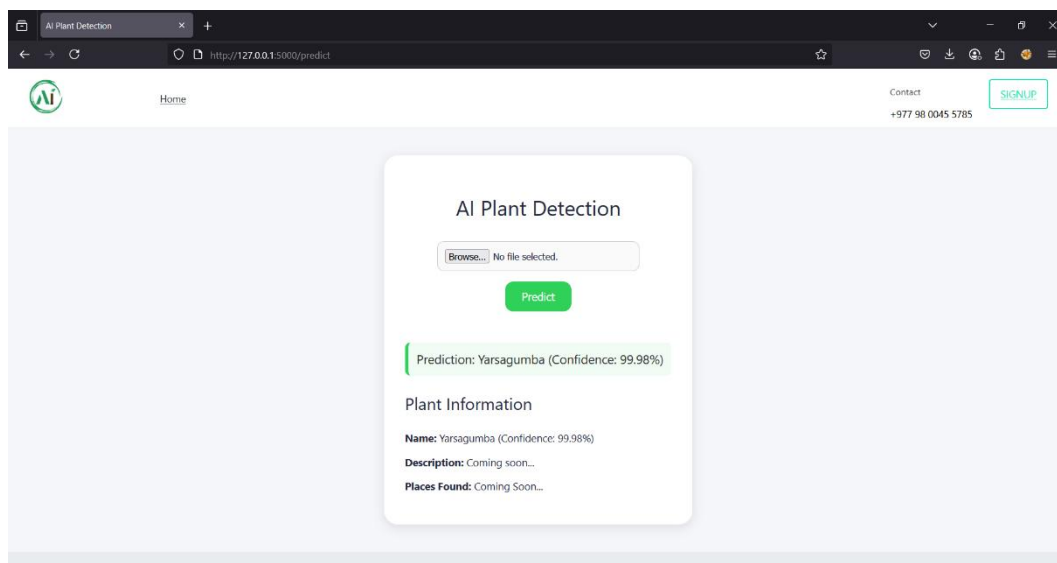
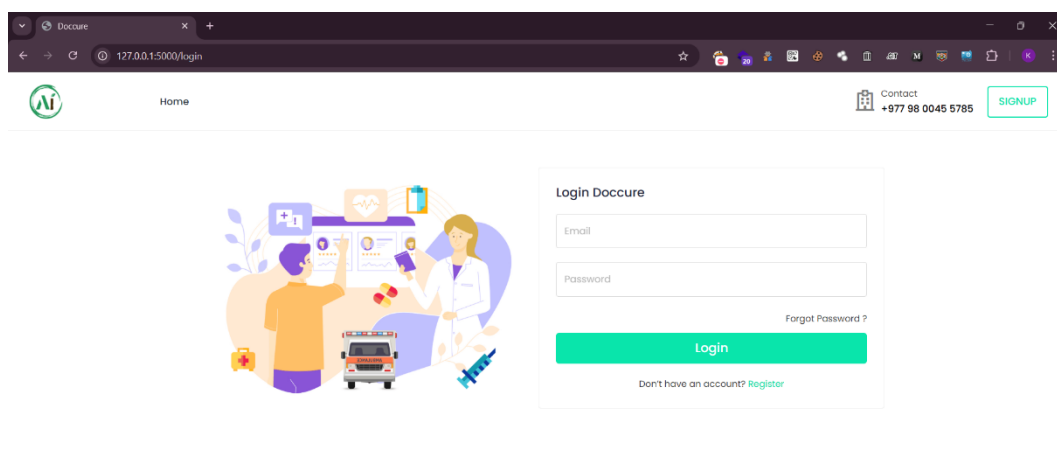


Figure 14: Image Identified

4.2.4 Doctor-Patient Login System

Secure login portals have been developed for both patients and Ayurvedic doctors. User roles are appropriately separated, and authenticated access is verified.



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Figure 15: Login Page

4.2.5 Homepage and Navigation

The homepage displays a clean and intuitive interface with easy access to the main features including chatbot, herb scanner, login options, and about section.

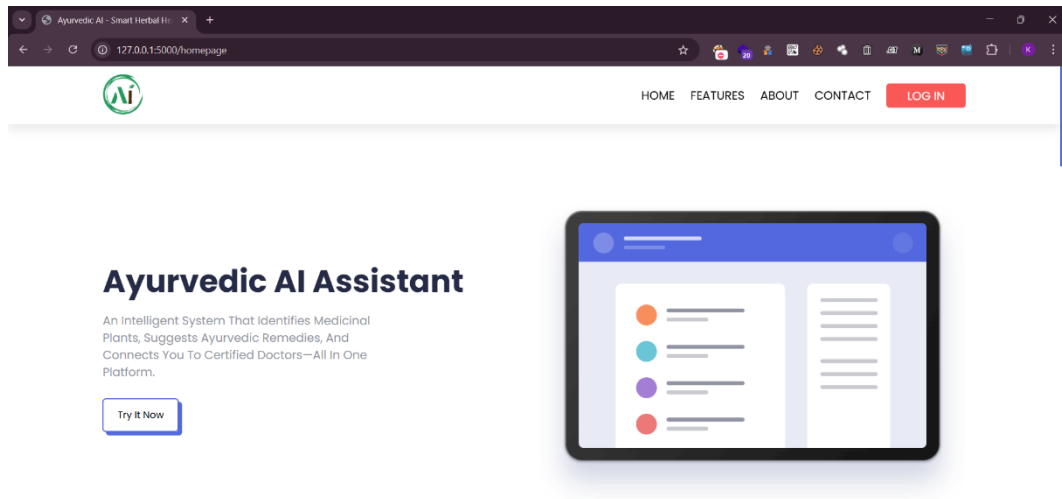


Figure 16: HomePage

4.3 Pending Tasks

The following tasks are remained to be completed.

- **Advanced Chatbot Development:** Enhancing the Chatbot to handle more complex queries and provide more accurate, context-aware responses.
- **Advanced Herb Image Classifier:** Improving the image classifier for higher accuracy and better recognition of diverse herb species.
- **"AICare Call" Feature Integration:** Adding the feature to enable real-time voice or video consultations within the system.
- **Performance Optimization:** Refining the system to run faster and more efficiently, reducing latency and resource usage.

Chapter 5: CHALLENGES AND RISK ANALYSIS

During the development of the herb image classifier, we faced challenges such as overfitting and under fitting when a model is too complex or trained too long on limited or noisy data, causing it to memorize rather than generalize. And when a model is too simple or insufficiently trained, failing to capture the underlying patterns in the data. These issues often caused misclassifications or failed predictions. Some of the main issues includes:

5. 1 Data-Related Issues:

During our project, we faced challenges with the dataset. Some classes had significantly fewer images, leading to imbalance and biased predictions. Additionally, a few mislabeled and low-quality images caused noise, which affected the model's learning accuracy.

5.2 Model-Related Issues:

Initially, we used a model that was either too simple or too complex for our data. This resulted in under fitting in the early stages, and later, overfitting when we used a deeper model without proper regularization. Tuning the architecture to fit our data took several iterations.

5.3 Training Issues:

Training was tricky due to hyperparameter tuning. At first, our learning rate and batch size were not optimal, slowing down convergence. We also didn't apply data augmentation early on, which limited the model's ability to generalize to new variations of images.

5.4 Risk Mitigation Strategies

To address the above technical and integration challenges, the following strategies were implemented:

- **Data Cleaning and Augmentation:** We thoroughly reviewed the dataset to remove mislabeled or low-quality images and applied various data augmentation techniques such as rotation, flipping, and scaling to increase diversity and balance class distribution.
- **Model Selection and Regularization:** After experimenting with different architectures, we selected a model that balanced complexity and performance. Regularization techniques like dropout and batch normalization were introduced to prevent overfitting.
- **Hyperparameter Tuning and Optimized Training:** We used grid search and experimentation to fine-tune hyperparameters like learning rate, batch size, and number of epochs. Early stopping and learning rate scheduling were also implemented to ensure stable and efficient training.
- **Iterative Testing and Feedback:** Throughout the project, we adopted an iterative development cycle, regularly testing the model on unseen data and incorporating feedback to refine both the model and the data pipeline.

Chapter 6: REMAINING WORK AND TIMELINE

6.1. Integration Plan for Classifier Herb Database

Although the herb image classifier's basic functionality has been implemented, the final integration with the backend herb database is still pending. This includes building a secure API bridge between the classifier and the backend, so that once a herb is identified, its complete profile including scientific name, medicinal uses, benefits, and regional info can be fetched and displayed in real time.

6.2. Final Testing and Debugging

A thorough testing phase is planned to ensure reliability, accuracy, and user experience. This will include:

- Unit Tests for backend APIs and data retrieval
- Integration Tests between image uploads, classification results, and database responses
- Simulated Use Cases, such as scanning real herb images and accessing Chatbot and AICare Call features under varied conditions

6.3. UI Polish and Error Handling

Some minor user interface issues remain, such as inconsistent layouts and lack of real-time loading indicators. Additionally, error handling will be enhanced for scenarios like:

- Unclear herb images
- Invalid or incomplete Chatbot inputs
- Server or database timeouts

6.4. Deployment and User Testing

The complete platform will be deployed to test server, where initial users (patients, students, and Ayurvedic practitioners) will be invited to interact with the app. This phase will include:

- Collecting real user feedback on accuracy, usability, and responsiveness
- Monitoring usage patterns and errors
- Refining features like the chatbot and the upcoming AICare Call before full-scale deployment.

Chapter 7: CONCLUSION

The Ayurvedic Intelligence project has made steady and meaningful progress toward creating a smarter way to connect people with traditional herbal medicine. At this point, we've successfully built the core structure of the system, including patient and doctor login, a basic herb image classifier, and an early version of the Chatbot. These features form the backbone of what we aim to offer an accessible platform that blends ancient Ayurvedic knowledge with modern AI tools.

One of the most exciting parts of this project is the ability to recognize herbs through images, making it easier for users to identify plants they may not be familiar with. Although this feature is still in its early stages, it already shows strong potential. We've also laid the groundwork for a feature called AICare Call, which will allow users to directly consult Ayurvedic experts, something we believe can make the platform even more useful in real-life health situations.

There's still work to do. We're currently focused on improving the image classifier so it can provide more detailed information about each herb, expanding the Chatbot's capabilities, and fine tuning the system overall. But the progress so far is encouraging, and we're confident that with continued effort, testing, and feedback, the platform can become a valuable tool for anyone interested in natural healing and Ayurveda.

In the long run, Ayurvedic Intelligence is more than just a tech project, it's a step toward preserving traditional knowledge while making it more available and relevant in today's digital world.

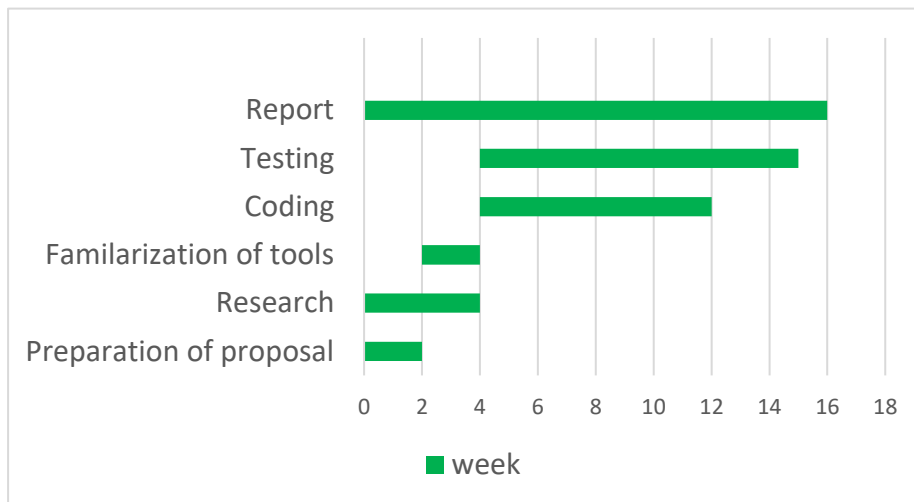
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APPENDICES

Gantt chart:

The Gantt chart shows the start and end dates of each task, as well as the dependencies between tasks.



Cost estimation:

The cost estimation includes a breakdown of the expected costs associated with the project, including labor costs, material costs, and other expenses.

Table 2: Cost Estimation

Development Activities	Status	Estimated Days	Estimated Hours/Day	Estimated Budget
Requirement Analysis	Complete	15	4	-
UI/UX Design	-	20	4	-
Coding	Complete	25	4	-
DB Designing	Complete	5	2	-
Testing	-	15	2	-

Report writing	-	27	4	Rs. 600
Total		112	-	Rs. 600